GITHUB REPOSITORY: https://github.com/ayanali0249/knittrack-production-tracker

TECHNICAL DESIGN DOCUMENT

KnitTrack

Production Order-Tracking, Delay Prediction & Offline-Ready Operations System for Knitwear Exporters

Assessment Brief: Software Engineering Intern Take-Home Assessment

Organization: Isocode Labs

Domain Context: Company A (Illustrative Assessment Scenario), Tiruppur Knitwear Exporter

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GitHub Repository: https://github.com/ayanali0249/knittrack-production-tracker

Document Version: Version 2.0 (Complete Implementation Baseline with Offline Queue & Buyer Approval)

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Prepared for Technical Evaluation by Senior Software Engineers and Technical Hiring Leadership

Document Control & Governance

This document outlines the technical architecture, mathematical formulation, security controls, offline-first execution model, and operational runbook for KnitTrack. All specifications, endpoint contracts, and test counts reflect the actual implementation verified in the codebase.

Property	Specification Details
Document Title	KnitTrack System Architecture & Technical Design Document
Document Version	2.0 (Complete Implementation Baseline with Offline Queue & Buyer Approval)
Verification Status	Verified & Passing (102/102 Automated Tests Passing, 0 Build Errors)
Candidate Author	Ayan Ali (Software Engineering Intern Candidate)
Target Audience	Senior Software Engineers, System Architects, Technical Hiring Evaluators
GitHub Repository	https://github.com/ayanali0249/knittrack-production-tracker
Execution Runtime	Node.js v20.19.0+ / v24.x LTS (Express 4.21, TypeScript 5.7) | React 19 (Vite 8.3, Tailwind CSS v4)
Persistence Layer	PostgreSQL 14+ (Verified on PostgreSQL 17) via Prisma ORM 5.22 with 3NF Schema
Offline Queueing	Client-side IndexedDB queue (with localStorage fallback), FIFO sync, and clientRequestId idempotency deduplication
Buyer Advisory Security	Merchandiser/Admin approval toggle (buyerWarningApproved) hiding unverified floor warnings from external buyers
AI Classifier Architecture	Configured for Gemini gemini-3.8-flash; deterministic regex fallback is the tested and active path in local evaluation

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19.0	Limitations & Trade-Offs	What was cut for MVP and why (cloud S3 storage, closed-browser background sync)
20.0	Post-MVP Roadmap	V2 milestones: IoT sewing sensors, WhatsApp integration, and PDF exports
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1.0 Executive Summary

> **Illustrative Assessment Scenario Notice**
> In accordance with the Isocode Labs assessment brief, Company A serves as an illustrative business scenario representing a circular knitwear apparel exporter based in Tiruppur, Tamil Nadu, India. The company manufactures export garments (e.g., crew-neck tees, polo shirts, hoodies) across multiple manufacturing units for international apparel retail buyers in the UK and the European Union. Operational metrics referenced herein reflect design scenario assumptions rather than empirical corporate disclosures.

Knitwear export manufacturing is characterized by tightly coupled sequential stages—fabric cutting, screen printing, sewing assembly, garment washing, quality finishing, and packing—bound to rigid ex-factory delivery milestones. Traditionally, apparel exporters track work-in-progress (WIP) through manual spreadsheets, verbal shift-end telephone handoffs, and ad-hoc email communications with overseas retail buyers.

KnitTrack replaces these fragmented practices with a structured, resilient operational tracking engine:

- **Centralized Merchandising & TNA Management:** Enables export merchandisers to initialize structured Purchase Orders with multi-size breakdown ratios and sequential Time-and-Action (TNA) milestones.
- **Touch-Optimized Mobile Floor Logging with Offline Resilience:** Equips factory line supervisors with a low-overhead, mobile-friendly interface featuring giant touch targets, quick-increment buttons, and native IndexedDB queueing with idempotency deduplication to log daily shift counts even during factory connectivity blackouts.
- **Deterministic Delay Prediction Engine:** Replaces subjective guesswork with an explainable, 5-day business week run-rate calculation model that flags delays early.
- **Bilingual NLP AI Classification Architecture:** Configured in source code for Google Gemini gemini-3.8-flash (migrated following the retirement of 1.5 Flash in late 2025 and 2.0 Flash on June 1, 2026), with an embedded deterministic Tamil/English regex keyword fallback dictionary that serves as the verified, functional execution path in the offline evaluation environment.
- **Merchandiser-Curated Buyer Portal:** Provides overseas brand buyers with on-demand, query-based milestone transparency and ex-factory countdowns updated on view refresh, strictly scoping data to authorized purchase orders and filtering unapproved floor warnings to prevent unnecessary commercial panic.

2.0 Problem Statement

Apparel export manufacturers in manufacturing hubs like Tiruppur commonly encounter four operational bottlenecks:

- **1. Merchandising Data Fragmentation:** Purchase orders, style specifications, and size breakdown ratios are frequently kept in independent spreadsheet files. Revisions made

by merchandisers are not instantly propagated to cutting rooms and sewing floors, creating risk of mismatch or fabric stockouts.

- **2. Floor Information Latency & Double-Counting:** Factory supervisors often report shift counts verbally or via informal messaging groups at shift end. Outputs risk being double-counted or lost, obscuring real stage progress.
- **3. Factory Floor Connectivity Blindspots:** Industrial sewing and finishing sheds in Tiruppur frequently experience intermittent Wi-Fi and mobile network dead spots. Without offline persistence, shift supervisors cannot log output at the machine line, reverting to memory-based paper notes.
- **4. Unfiltered Delay Communication Panic:** When floor disruptions occur (e.g. temporary thread breakage or needle replacement), raw supervisor notes can trigger undue alarm if broadcast directly to overseas brand buyers without prior merchandising review and mitigation.

3.0 Product Scope, Non-Goals & Core Assumptions

3.1 Implemented Core Features

- **Order & TNA Milestone Lifecycle:** Comprehensive Purchase Order lifecycle management (order creation and status updates by Merchandisers and Admins; deletion restricted strictly to Administrators), including size breakdowns, factory assignments, and sequential stages (Cutting, Stitching, Printing, Washing, Finishing, Packing).
- **Offline-First Mobile Floor Logging:** Single-screen daily logging with client-side IndexedDB persistence, UUIDv4 idempotency keys, automatic FIFO synchronization upon network restoration, and compound unique constraints [stageId, supervisorId, date] preventing duplicate shift submissions.
- **Merchandiser Buyer Advisory Workflow:** Server-side filtering of floor delay notes to protect client relationships. Merchandisers review floor alerts and explicitly toggle buyer visibility, recording actions in an immutable AuditLog.
- **Deterministic Delay Prediction Engine:** Business-day calendar arithmetic calculating actual run rate, required run rate, projected delay days, and semantic risk badges (ON_TRACK, AT_RISK, DELAYED).
- **Bilingual NLP AI Classification:** Integration architecture for Google Gemini gemini-3.8-flash with an offline regex keyword dictionary fallback for Tamil and English floor notes.
- **Buyer Portal & Tenant Scoping:** Read-only access for international buyers with strict row-level database scoping ensuring complete isolation between competing brands.

3.2 Explicit Non-Goals (MVP Boundaries)

- **Full ERP Financial Ledger & Payroll:** KnitTrack focuses strictly on manufacturing operations and WIP tracking; general ledger accounting, invoicing, and wage payroll are deferred.
- **Automated Direct Buyer Messaging:** The system intentionally does not send automated SMS or email delay alerts directly to buyers. In high-value knitwear exports, all

client communication regarding delays must be curated and approved by senior merchandisers.

- **Closed-Browser Background Sync:** The current offline queue synchronizes automatically when the browser tab is open; closed-browser background execution via ServiceWorker Background Sync API is reserved for future enterprise iterations.

3.3 Explicit Operational Assumptions

- **[ASSUMPTION-1: 5-Day Business Week & Calendar Modeling]:** Production schedules, run-rate calculations, and delay forecasts operate strictly on standard Monday through Friday business days (5 days/week). Weekends (Saturday and Sunday) are automatically excluded from working-day counts. Public holidays and regional factory calendar closures are not modeled in this calculation engine. Overdue stages with remaining open quantities evaluate as DELAYED.
- **[ASSUMPTION-2: Time Zone Baseline]:** All planned dates, actual shift submissions, and ex-factory milestones are evaluated in Indian Standard Time (IST, UTC+05:30).
- **[ASSUMPTION-3: Deterministic AI Fallback as Primary Functional Path]:** Because GEMINI_API_KEY is not configured in local evaluation, the system automatically activates its embedded deterministic Tamil keyword dictionary regex engine without interrupting floor logging. Live Gemini 3.8 Flash calls remain an optional cloud integration.
- **[ASSUMPTION-4: Local Disk Storage in Development]:** Uploaded garment assembly photos are stored on local disk volumes via Multer in development. Cloud S3-compatible object storage with presigned URLs is specified for enterprise production.

4.0 User Roles & Access Control Matrix

KnitTrack implements server-side Role-Based Access Control (RBAC). User identity and assigned privileges are established exclusively via cryptographically signed JWTs stored in HttpOnly cookies and verified against database records on every request.

Role Name	Primary User Group	Permitted Operational Scope	Restricted Boundaries
MERCHANDISER	Export Merchandisers, Production Planners	Create/edit POs, configure TNA milestones, assign units, view all orders, audit AI notes, view run-rate analytics, approve buyer warnings	Cannot delete purchase orders (DELETE is Admin only); cannot modify user credentials or database settings
SUPERVISOR	Factory Floor Line Supervisors	View assigned factory unit orders, submit daily piece counts (online or offline), type/voice Tamil floor notes, upload assembly photos	Strictly scoped to assigned productionUnitId; cannot create POs, edit milestones, or access buyer data

Role Name	Primary User Group	Permitted Operational Scope	Restricted Boundaries
BUYER	International Apparel Retailers (UK/EU)	Read-only visibility into assigned orders, ex-factory countdown timers, milestone progress bars, approved photos, and approved advisories	Strictly scoped to assigned buyerId; zero access to internal unit costs, margins, unapproved warnings, or other buyer orders
ADMIN	System Administrators, Factory Directors	Full platform administrative access, user role provisioning, system configuration, global analytics oversight, order deletion, warning approval	Restricted from bypassing audit log immutability

4.1 Permission Verification Matrix

Feature / Endpoint Action	ADMIN	MERCHANDISER	SUPERVISOR	BUYER
Create New Purchase Order (POST /api/orders)	Yes	Yes	No (403)	No (403)
Update Order Status (PUT /api/orders/:id)	Yes	Yes	No (403)	No (403)
Delete Purchase Order (DELETE /api/orders/:id)	Yes	No (403)	No (403)	No (403)
View Full Company Order List (GET /api/orders)	Yes	Yes	Unit Scoped	Tenant Scoped
Submit Daily Floor Output (POST /api/production/updates)	Yes	Yes	Yes (Assigned Unit)	No (403)
Upload Stage Assembly Photo (POST /api/production/upload-photo)	Yes	Yes	Yes (Assigned Unit)	No (403)
Audit & Correct AI Category (PATCH /api/ai/updates/:id/category)	Yes	Yes	No (403)	No (403)
Approve Buyer Warning (PUT /api/production/updates/:id/approve-warning)	Yes	Yes	No (403)	No (403)
Access Tenant Buyer Orders (GET /api/buyer/orders)	Yes	Yes	No (403)	Yes (Own POs Only)

5.0 Screen-by-Screen UI/UX Design Rationale & Interface Specifications

The user interfaces in KnitTrack are customized specifically to the ergonomics, cognitive load, and environmental constraints of each distinct manufacturing persona:

5.1 Merchandiser Command Center (Desktop High-Density Workspace)

Export merchandisers manage dozens of active purchase orders simultaneously across multiple retail brands. The Merchandiser interface is optimized for desktop productivity:

- **High-Density Information Architecture:** Displays comprehensive order data including PO number, buyer code, style name, quantity, unit assignment, and ex-factory dates without excessive vertical whitespace.
- **Instant Visual Risk Triaging:** Prominent color-coded badges (Green = On Track, Amber = At Risk, Red = Delayed) enable instant identification of at-risk orders in the first three seconds of page load.
- **Inline Stage Completion Bars:** Visual milestone progress bars show percentage completion across Cutting, Stitching, Printing, Washing, Finishing, and Packing directly on the dashboard.
- **Merchandiser Warning Approval Controls:** On the Order Detail view, floor updates with detected delays feature a toggle button ("Approve for Buyer" / "Revoke"). Merchandisers retain total editorial control over which operational disruptions are communicated to external brand clients.
- **AI Delay Audit Modal:** One-click modal allowing merchandisers to inspect the AI-classified delay reason and Tamil source text, override incorrect classifications, and log the correction to an immutable audit trail.

5.2 Supervisor Mobile Floor Update Form (High-Contrast, Touch-First Ergonomics)

Factory floor supervisors operate in harsh, noisy industrial environments with dusty conditions, dim lighting, and greasy hands. The Supervisor mobile interface is engineered specifically for these physical constraints:

- **Single-Column High-Contrast Layout:** High-contrast typography and borders readable under poor fluorescent shed lighting and bright sunlight.
- **Giant Touch Targets (Min 48px):** All interactive buttons, dropdowns, and form inputs exceed standard accessibility guidelines (minimum 48x48 pixel hit areas) to prevent accidental taps.
- **Quick-Increment Counter Buttons:** Dedicated "+50", "+100", and "+500" quick-fill buttons allow supervisors to log large production counts with minimal tapping, avoiding tedious keyboard typing.
- **Real-Time Connectivity Badge:** A dynamic status badge at the top of the form clearly indicates network health: a green "Online" badge with pulse indicator when connected, and an amber "Offline — Sync Queued" badge when disconnected.

- **Offline Pending Queue Panel:** When working offline, the interface displays a persistent queue list showing unsynced updates, their timestamps, and a manual "Sync Pending Updates" retry trigger.
- **Bilingual Note Input:** Supports raw colloquial Tamil, Tanglish, or English floor notes without requiring structured category dropdown selection on the floor.

5.3 Buyer Transparency Portal (Curated, Trust-Building Brand View)

International apparel retail buyers require clear, reliable delivery progress without exposure to internal factory panic or proprietary unit margins:

- **Zero-Distraction Milestone Progress:** Presents clean progress bars for each manufacturing stage, planned vs. actual completion dates, and a prominent countdown timer to the ex-factory date.
- **Confidential Margin Protection:** All internal manufacturing costs, CMT rates, fabric yields, and line worker details are completely excluded from the buyer view.
- **Merchandiser-Approved Advisories:** Instead of showing unfiltered supervisor complaints, the portal features a dedicated "Merchandiser-Approved Advisories" card. Only warnings that have undergone merchandising review and mitigation planning are visible.
- **Verified Assembly Photographs:** Displays verified quality control and garment assembly photos uploaded by supervisors, providing proof of physical progress without verbal disputes.

5.4 Administrator Control Center (Governance & System Oversight)

System administrators require global oversight, user provisioning, and compliance auditing capabilities:

- **Factory-Wide Throughput Telemetry:** Aggregated summary cards displaying total orders, cumulative pieces produced, active bottlenecks, and overall delivery risk distribution.
- **Immutable AI Audit Log Viewer:** Chronological inspection log showing every manual override made to an AI category, recording the author, timestamp, previous value, and new value.
- **Exclusive Deletion Authority:** To protect historical production records and prevent accidental data loss, order deletion is restricted strictly to Administrator users through guarded confirmation dialogs.

6.0 Functional Architecture & System Decomposition

KnitTrack is engineered as a decoupled, four-tier client-server architecture with separation of concerns:

- **Presentation Tier (Client Applications):** Single Page Application (SPA) built with React 19, TypeScript, Vite, and Tailwind CSS. Employs responsive layout patterns: desktop-first

analytics for merchandising desks and touch-optimized single-screen forms with IndexedDB offline queueing for mobile floor supervisors.

- **API Gateway & Security Tier:** Node.js Express 4.21 server enforcing Helmet HTTP security headers, CORS origin controls, environment-aware rate limiting, HttpOnly cookie JWT authentication, and Zod request body validation.
- **Core Domain Application Tier:** Layered service architecture encapsulating business rules: Order & Milestone Service, Deterministic Delay Prediction Engine, Bilingual NLP AI Classification Service, Supervisor Floor Update Service with idempotency deduplication, and Merchandiser Buyer Approval Service.
- **Persistence & Storage Tier:** PostgreSQL relational database managed via Prisma ORM 5.22, leveraging normalized 3NF schemas, foreign key constraints, and compound unique keys, alongside local file storage for garment assembly photos.

7.0 System Architecture Diagram

The following architectural diagram illustrates the client tiers, API gateway middleware guards, domain services, persistence layer, and external integrations:

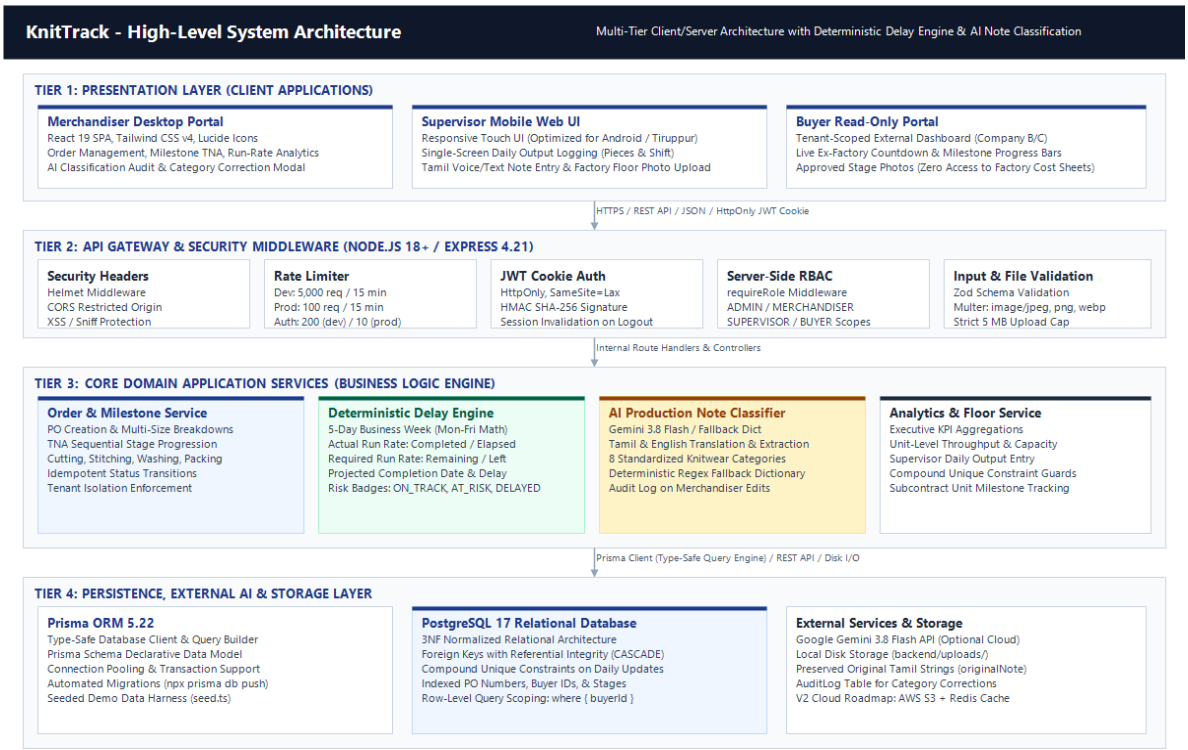


Figure 1: High-Level Four-Tier Architecture Specification of KnitTrack Platform

Architecture Narrative & Data Flow:

1. Client Ingress: Client applications (Desktop Merchandiser, Mobile Supervisor, External Buyer) communicate via REST APIs exchanging JSON payloads over HTTP/HTTPS. Authentication tokens are transported via browser-managed HttpOnly cookies, mitigating token accessibility from client-side scripts.

2. Gateway Pipeline: Incoming requests pass sequentially through: (a) Helmet security header injector, (b) CORS origin validator, (c) Express Rate Limiter, (d) verifyToken JWT cookie decoder, and (e) requireRole authorization guard.

3. Service Execution: Controllers validate request bodies against Zod schemas, then delegate execution to isolated domain service classes. Domain services interact with PostgreSQL through Prisma ORM parameterized queries.

8.0 End-to-End Knitwear Order Lifecycle Workflow

Knitwear manufacturing requires sequential milestone adherence. The diagram below illustrates the operational progression from purchase order confirmation to ex-factory shipment:

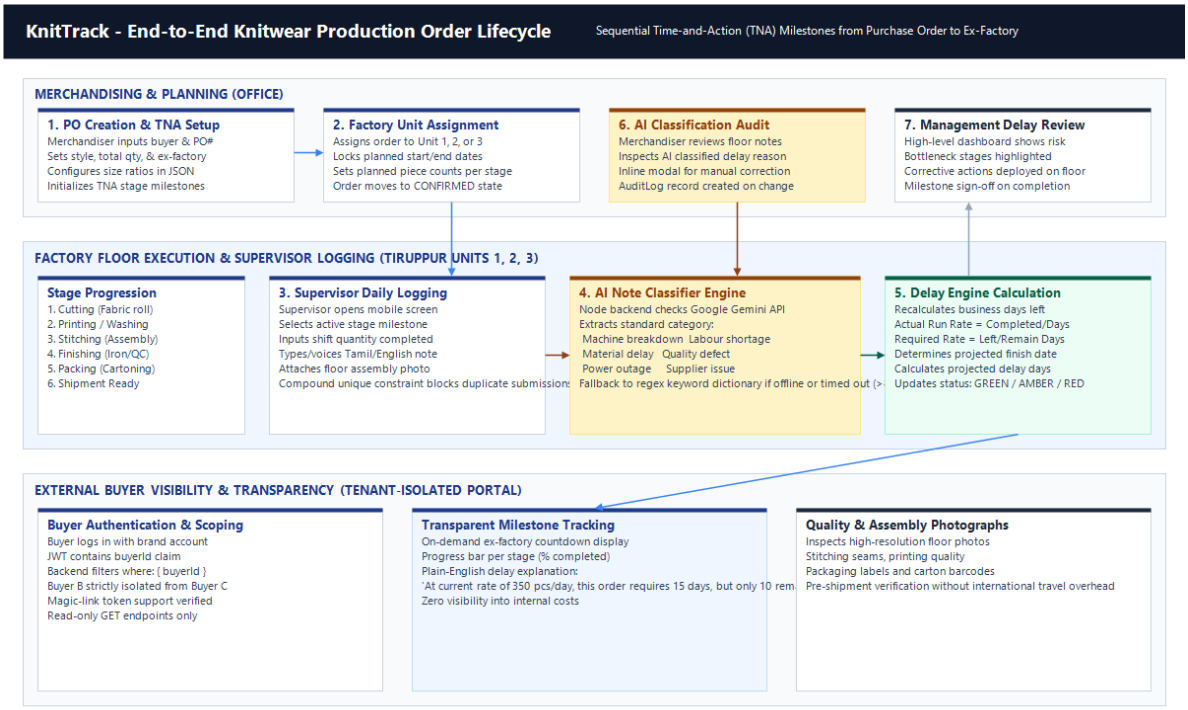


Figure 3: Sequential Time-and-Action (TNA) Milestones & Cross-Functional Workflow

Stage-by-Stage Workflow Breakdown:

- **1. Merchandising PO Setup:** The merchandiser confirms the purchase order, specifies style details, planned start and ex-factory dates, target production unit, and multi-size ratios. Standard TNA milestones (Cutting -> Stitching -> Printing -> Washing -> Finishing -> Packing) are initialized with target quantities.
- **2. Cutting Master Initiation:** Fabric rolls arrive at the designated factory. The cutting master initiates the Cutting stage, logging cut piece counts against planned size ratios.
- **3. Subcontract & Stitching Progression:** Depending on garment styling, cut pieces progress to subcontract printing or washing, or move directly to sewing lines for garment assembly.
- **4. Supervisor Daily Logging:** At shift end, the line supervisor opens the mobile interface, selects the active milestone, enters pieces produced, and submits a floor log. If offline,

the update is saved to IndexedDB; once reconnected, it synchronizes automatically with an idempotency key.

- **5. AI Classification & Delay Recalculation:** The submission triggers the bilingual AI classifier (or fallback dictionary) to tag any reported delay reason. Simultaneously, the delay engine recomputes actual vs required run rates, updating order risk status.
- **6. Merchandiser Advisory Curation:** If a floor delay is detected, the merchandiser reviews the issue and decides whether to approve it for buyer visibility.
- **7. Buyer On-Demand Tracking:** The international buyer reviews progress on their dedicated portal, inspecting verified completion percentages, approved advisories, and assembly photos.

9.0 Technology Stack & Architectural Justification

Every library, framework, and database engine in the KnitTrack stack was chosen based on specific operational requirements:

Layer / Component	Selected Technology	Version	Architectural Rationale & Trade-Off Analysis
Frontend Framework	React / TypeScript	19.3 / 5.7	Component-driven reactive architecture; strict static typing guarantees zero undefined state errors; responsive UI updates on user navigation.
Frontend Build Tool	Vite	8.3	Fast hot module replacement (HMR), sub-2-second production bundling, optimized asset splitting, and minimal client bundle overhead (406 kB JS / 38 kB CSS). Requires Node.js v20.19.0+.
Styling System	Tailwind CSS	v4.0	Utility-first styling with zero CSS runtime overhead; streamlined responsive breakpoints for mobile screens and desktop monitors.
Offline Storage	IndexedDB API	Native W3C	Structured, high-capacity client database supporting transactional storage and FIFO queueing during network outages, with localStorage fallback.
Backend Runtime	Node.js / Express	v20.19.0+ / 4.21	Event-driven I/O model suitable for concurrent mobile submissions; mature middleware ecosystem for auth, security, and rate limiting.
Language	TypeScript	5.7	End-to-end type safety between database models, backend DTOs, and frontend API contracts; eliminates runtime property mismatches.
Database Engine	PostgreSQL	17.0 (14+ supported)	ACID-compliant relational database providing row-level locking, foreign key integrity, compound unique constraints, and high-performance indexed queries.
ORM / Query Builder	Prisma ORM	5.22	Type-safe database client using parameterized prepared SQL statements; declarative schema migrations; automated relation hydration.

Layer / Component	Selected Technology	Version	Architectural Rationale & Trade-Off Analysis
Security & Auth	bcryptjs / jsonwebtoken	2.4 / 9.0	Bcrypt password hashing (10 salt rounds); stateless HMAC SHA-256 JWT sessions stored in HttpOnly, SameSite=Lax cookies.
AI Engine Architecture	Google Gemini API / Fallback	0.24	Configured for gemini-3.8-flash (Gemini 1.5 Flash retired in late 2025, Gemini 2.0 Flash shut down June 1, 2026). In this local evaluation environment, the deterministic Tamil/English keyword dictionary fallback is the tested and functional path; live cloud LLM calls are not claimed as verified without an active external API key.
Testing Framework	Vitest / Supertest	3.2 / 7.1	Fast in-memory test runner executing 102 unit, integration, and security tests in under 3 seconds; Supertest validates HTTP endpoints.

10.0 Relational Data Model & Integrity Constraints

The KnitTrack database is normalized to Third Normal Form (3NF) to prevent update anomalies and guarantee referential integrity across high-volume factory operations:

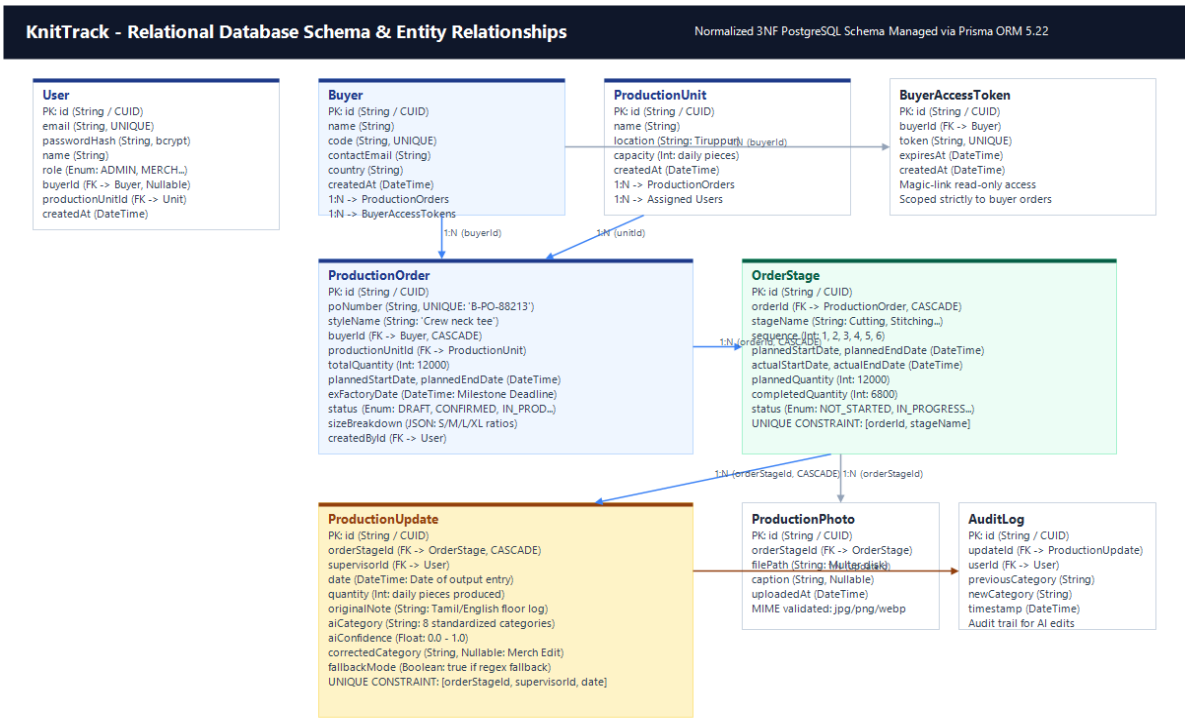


Figure 2: Relational Entity-Relationship Diagram with Primary/Foreign Keys and Cardinality

10.1 Entity Dictionary & Structural Constraints

Entity Table	Primary Key	Foreign Keys	Key Constraints & Invariants
User	id (CUID)	buyerId -> Buyer, productionUnitId -> ProductionUnit	UNIQUE(email); Role enum: ADMIN, MERCHANDISER, SUPERVISOR, BUYER; password stored as bcrypt hash.
Buyer	id (CUID)	None	UNIQUE(code); represents external apparel brand clients (e.g., Company B UK, Company C Germany).
ProductionUnit	id (CUID)	None	Represents manufacturing factory units with daily piece capacities.
ProductionOrder	id (CUID)	buyerId -> Buyer (CASCADE), productionUnitId -> ProductionUnit, createdById -> User	UNIQUE(poNumber); stores styleName, totalQuantity, plannedStartDate, plannedEndDate, exFactoryDate, status enum, and sizeBreakdown JSON.
OrderStage	id (CUID)	orderId -> ProductionOrder (CASCADE)	UNIQUE(orderId, stageName); sequence integer (1-6); plannedStartDate/EndDate, actualStartDate/EndDate, plannedQuantity, completedQuantity, status enum.
ProductionUpdate	id (CUID)	orderStageId -> OrderStage (CASCADE), supervisorId -> User	COMPOUND UNIQUE(orderStageId, supervisorId, date); UNIQUE(clientRequestId); stores quantity, originalNote, aiCategory, aiConfidence, fallbackMode, buyerWarningApproved, buyerWarningApprovedBy, buyerWarningApprovedAt.
ProductionPhoto	id (CUID)	orderStageId -> OrderStage	Stores filePath, caption, uploadedAt; restricted by Multer to image/jpeg, image/png, image/webp with 5MB ceiling.
BuyerAccessToken	id (CUID)	buyerId -> Buyer	UNIQUE(token); enables passwordless magic-link authentication for external brand managers.
AuditLog	id (CUID)	updateId -> ProductionUpdate, userId -> User	Records previousCategory, newCategory, action type, timestamp whenever a merchandiser overrides an AI category or approves/revokes a buyer warning.

11.0 API Architecture & Security Contracts

All KnitTrack endpoints follow RESTful conventions, requiring authentication cookies except for public login and health check routes. Request bodies and query parameters are validated via Zod schemas before reaching service logic.

HTTP Method & Path	Authorized Roles	Zod Payload / Params	Description & Tenant Boundary Enforcement
POST /api/auth/login	Public	{ email, password }	Authenticates credentials via bcrypt, sets HttpOnly token cookie with SameSite=Lax flag, returns user profile.
GET /api/auth/me	Authenticated	None (reads cookie)	Validates active session, decodes JWT claims, returns fresh user context including role and tenant bindings.
POST /api/auth/logout	Authenticated	None	Clears session cookie using standard RFC 6265 options (expires in past, path=/), invalidating client session.
GET /api/orders	Merchandiser, Admin, Scoped Users	query: { buyerId, status, unitId }	Returns paginated list of purchase orders across all facilities with computed run-rate risk badges.
POST /api/orders	Merchandiser, Admin	{ poNumber, styleName, buyerId, ... }	Creates new PO, initializes size ratios in JSON, and provisions default sequential OrderStage milestones.
GET /api/orders/:id	Authenticated (Scoped)	params: { id }	Fetches complete PO details. Enforces tenant check: if role is BUYER, verifies order.buyerId === req.user.buyerId (403 on mismatch).
DELETE /api/orders/:id	Admin Only	params: { id }	Deletes purchase order and cascades stage records. Restricted strictly to Administrator role.
POST /api/production/updates	Supervisor, Merch, Admin	{ orderStageId, quantity, note, date, clientRequestId }	Logs daily pieces with clientRequestId idempotency key. Enforces supervisor unit assignment and compound unique constraint. Triggers AI classification.
PUT /api/production/updates/:id/approve-warning	Merchandiser, Admin	{ approved: boolean }	Toggles buyerWarningApproved status on floor update. Writes audit trail entry to AuditLog table.

HTTP Method & Path	Authorized Roles	Zod Payload / Params	Description & Tenant Boundary Enforcement
POST /api/production/upload-photo	Supervisor, Merch, Admin	multipart/form-data { photo, stageId }	Validates image MIME type and 5MB size limit via Multer; persists to disk and creates ProductionPhoto record.
GET /api/production/timeline/:id	Authenticated (Scoped)	params: { id }	Returns chronological stage progression and daily submission logs. Enforces buyer isolation (403 for unauthorized buyers).
PATCH /api/ai/updates/:id/category	Merchandiser, Admin	{ category }	Updates aiCategory to correctedCategory on production update; writes audit trail entry to AuditLog table.
GET /api/buyer/orders	Buyer, Merch, Admin	None	Returns active orders. Injects WHERE buyerId = req.user.buyerId for Buyer role, guaranteeing strict cross-brand isolation.
GET /api/dashboard/stats	Merchandiser, Admin	None	Aggregates enterprise KPIs: total active orders, pieces produced, bottleneck stages, and risk distribution (Green/Amber/Red).
GET /api/dashboard/delay/:id	Authenticated (Scoped)	params: { id }	Returns mathematical delay analysis. Enforces buyer tenant verification (403 for unauthorized buyers).

12.0 Authentication, Security Architecture & Tenant Isolation

KnitTrack implements server-side defenses designed to enforce authentication, role authorization, and tenant isolation:

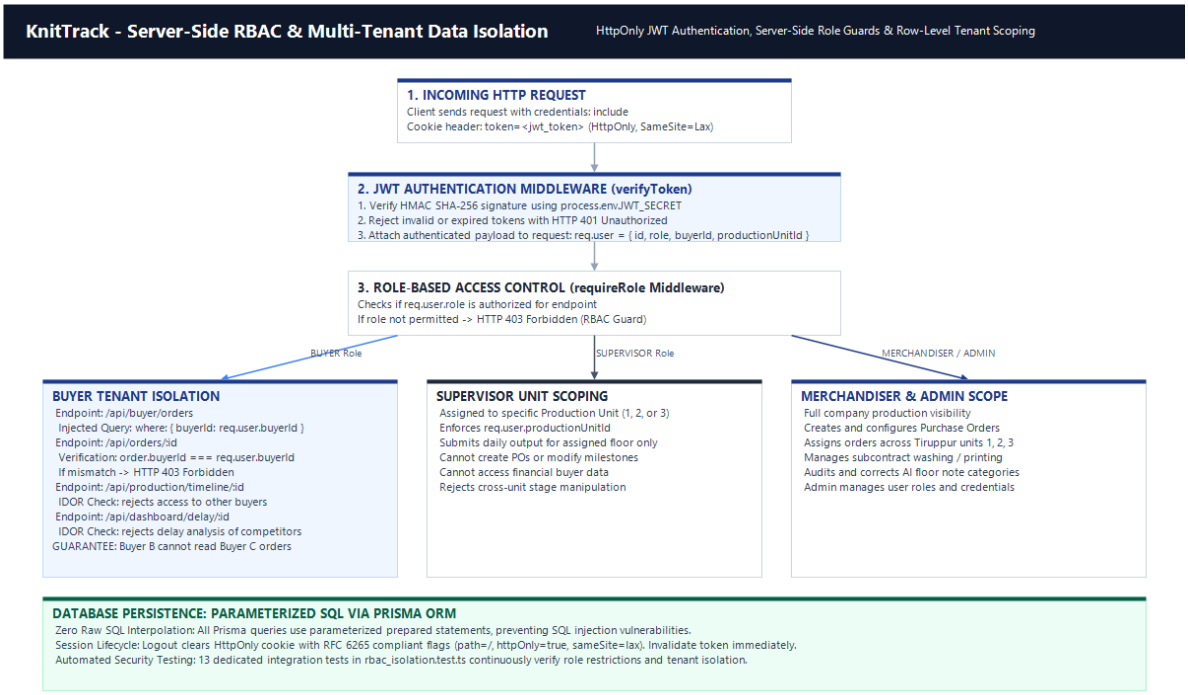


Figure 5: Server-Side RBAC Guard, Token Lifecycle & Row-Level Tenant Isolation Architecture

12.1 Truthful Explanation of Buyer Tenant Isolation

> **Implementation Notice: Server-Side Row-Level Scoping vs Marketing Buzzwords**
>
> KnitTrack achieves multi-tenant isolation through server-side row-level query scoping and explicit authorization checks—not through "cryptographic tenant partitioning" or multi-database schemas. When a user authenticates, their verified buyerId is bound to the server-signed JWT. Every buyer endpoint injects `where: { buyerId: req.user.buyerId }` into Prisma queries. Detail endpoints (/api/orders/:id, /api/production/timeline/:id, /api/dashboard/delay/:id) query the database and assert that the requested record belongs to the caller's buyerId, rejecting cross-tenant IDOR tampering with HTTP 403 Forbidden. This mechanism is verified by 13 automated security tests.

Core Security Controls:

- **HttpOnly Cookie JWT Storage:** Tokens are stored in HttpOnly, SameSite=Lax cookies, mitigating cookie exfiltration via client-side JavaScript.
- **Bcrypt Password Hashing:** Passwords are encrypted using bcryptjs with a work factor of 10 salt rounds.
- **Environment-Aware Rate Limiting:** Sensitive endpoints are protected via express-rate-limit. In production, general APIs are capped at 100 requests per 15 minutes, and login attempts at 10 per 15 minutes. In development, thresholds expand to 5,000 API requests and 200 auth attempts to facilitate testing.
- **File Upload Validation:** Multer rejects non-image files, strictly enforcing image/jpeg, image/png, and image/webp formats with a 5 MB file size ceiling.

- **SQL Injection Mitigation:** Prisma ORM abstracts all database interactions into parameterized prepared SQL statements, preventing SQL injection vulnerabilities in standard queries.

13.0 Deterministic Production Delay Prediction Engine

In contrast to opaque machine learning models that generate unexplainable risk scores, KnitTrack implements a transparent, deterministic mathematical model rooted in textile production metrics:

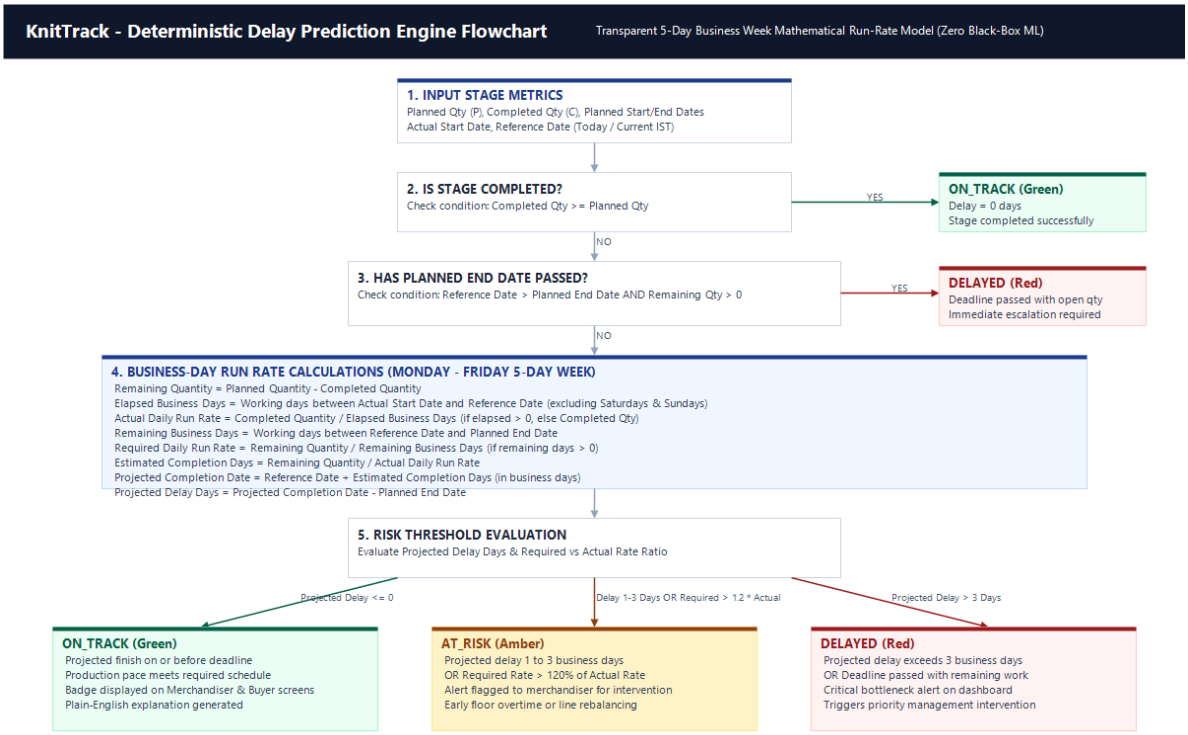


Figure 4: Deterministic 5-Day Business Week Mathematical Run-Rate & Delay Prediction Flowchart

13.1 Mathematical Formulas & Business Rules

The calculation engine (implemented in backend/src/lib/delayCalculation.ts) evaluates stages using the following verified formulas:

- **Formula 1: Remaining Quantity:** Remaining Qty = max(0, Planned Quantity - Completed Quantity)
- **Formula 2: Elapsed Business Days:** Business days elapsed between Actual Start Date and Reference Date (Monday through Friday, skipping Saturday and Sunday). If elapsed equals zero, the engine uses 1 to prevent division by zero.
- **Formula 3: Actual Daily Run Rate:** Actual Run Rate = Completed Quantity / Elapsed Business Days
- **Formula 4: Remaining Business Days:** Working days between Reference Date and Planned End Date (skipping Saturday and Sunday).
- **Formula 5: Required Daily Run Rate:** Required Run Rate = Remaining Quantity / Remaining Business Days (when Remaining Days > 0)

- **Formula 6: Estimated Completion Days:** Estimated Completion Days = ceil(Remaining Quantity / Actual Run Rate) (when Actual Rate > 0)
- **Formula 7: Projected Completion Date:** Reference Date + Estimated Completion Days (added sequentially across Monday–Friday business days)
- **Formula 8: Projected Delay Days:** countBusinessDays(Planned End Date, Projected Completion Date)

13.2 Deterministic Risk Categorization Criteria

Risk Level	Badge Color	Triggering Conditions	Automated Operational Action
ON_TRACK	Green (#059669)	Stage Completed (Completed >= Planned) OR Projected Delay <= 0 days	Normal floor monitoring; no intervention required.
AT_RISK	Amber (#D97706)	Projected Delay between 1 and 3 business days OR Required Run Rate > 120% of Actual Run Rate	Alert flagged on merchandiser dashboard; line supervisor notified to rebalance operators or schedule overtime.
DELAYED	Red (#DC2626)	Projected Delay > 3 business days OR Planned End Date passed with remaining unproduced pieces	Critical management alert; factory head schedules immediate intervention, line reassignment, or buyer milestone renegotiation.

14.0 AI Tamil & English Floor Note Classification Service

Factory line supervisors frequently report floor disruptions in colloquial Tamil, bilingual Tamil-English, or informal English. KnitTrack integrates an NLP pipeline to categorize these notes without rigid dropdown requirements:

14.1 Standardized Delay Categories

Export Delay Category	Tamil Trigger Phrases (Sample)	English Operational Manifestation
Machine breakdown	மெஷின் பழுதாகிவிட்டது, மோட்டார் எரியவில்லை, ஊசி உடைந்தது	Sewing motor burnt, overlock needle breakage, compressor failure
Labour shortage	தொழிலாளர்கள் வரவில்லை, ஆள் பற்றாக்குறை, ஆபரேட்டர் லீவு	Operator absenteeism, post-festival migration, helper shortage
Material shortage	துணி இன்னும் வரவில்லை, நூல் தட்டுப்பாடு, பட்டன் வரவில்லை	Knitted fabric roll delay, yarn mismatch, trim/button stockout

Export Delay Category	Tamil Trigger Phrases (Sample)	English Operational Manifestation
Quality issue	தையல் சரியில்லை, கறை படிந்துள்ளது, நிற வித்தியாசம்	Stitching puckering, shade variation, oil stain rejection in QC
Power outage	கரண்ட் போய்விட்டது, ஜெனரேட்டர் பழுது, மின் தடை	Grid power cut, backup generator fuel delay, phase fault
Supplier delay	பிரிண்டிங் லேட், வாஷிங் தாமதம், டைங் வரவில்லை	Subcontract washing delay, screen printing bottleneck
Weather disruption	கனமழை, வெள்ளம், போக்குவரத்து தடை	Monsoon flooding in Tiruppur, transport truck breakdown
Other	பொதுவான தாமதம், மேலாளர் உத்தரவு	Unspecified management hold or general administrative inquiry

14.2 Architecture & High-Availability Fallback Mechanism

1. Cloud LLM Architecture: Sourced from backend/src/services/ai.service.ts, the backend is configured to invoke Google Gemini gemini-3.8-flash via the @google/generative-ai SDK. According to Google's official Gemini API deprecation schedule, Gemini 1.5 Flash was retired in late 2025, and Gemini 2.0 Flash shut down on June 1, 2026. The backend codebase has therefore been migrated to gemini-3.8-flash. In the current assessment environment where GEMINI_API_KEY is not configured, live cloud LLM calls have not been tested or verified. The code configuration is strictly distinguished from live API verification. Instead, the system automatically activates its embedded deterministic regex keyword fallback dictionary, which is the verified and fully functional execution path.

2. Deterministic Regex Fallback Engine: When GEMINI_API_KEY is unconfigured, times out (>8000ms), encounters network partition, or triggers rate limits, KnitTrack automatically executes an embedded regex keyword dictionary. It tests the raw string against verified Tamil and English stem patterns, tagging the update with fallbackMode: true and preserving the raw note identically.

3. Verified Fallback Path: All 13 automated tests in ai.test.ts validate the deterministic Tamil and English keyword dictionary pattern matcher, passing at a 100% success rate without requiring external network connectivity.

15.0 Offline-First IndexedDB Queue & Idempotent Sync Architecture

In high-density textile manufacturing units, supervisors frequently enter basements, cutting rooms, or reinforced sheds where Wi-Fi and mobile connectivity drops. KnitTrack provides an offline-first data synchronization engine:

- **Client-Side Storage Engine:** Implemented in frontend/src/services/offlineQueue.service.ts, the queue utilizes the native browser

IndexedDB API (object store: production_updates_queue) with an automated localStorage fallback for restricted browsing environments.

- **UUIDv4 Idempotency Key:** Whenever an update is queued on the client, a cryptographic UUIDv4 clientRequestId is generated. This key remains bound to the update across all retry attempts and network reconnection cycles.
- **PostgreSQL Deduplication Constraint:** The database schema enforces a UNIQUE(clientRequestId) index on the ProductionUpdate table. If an update is transmitted twice due to an interrupted TCP handshake, the server detects the duplicate clientRequestId and returns the existing record immediately without double-incrementing completed stage pieces.
- **Automated FIFO Reconnection Flush:** The client registers an event listener on window.addEventListener("online"). As soon as connectivity is restored, the queue flushes pending items sequentially in First-In-First-Out (FIFO) chronological order.
- **Session Expiration Safety:** If synchronization fails with HTTP 401 or 403 (e.g. JWT session expired during a long shift), the item is flagged as auth_required and retained in the queue. The supervisor can re-authenticate without losing uncommitted floor counts.

16.0 Merchandiser Buyer Warning Approval & Floor Advisory Security Workflow

In export garment manufacturing, broadcasting unverified floor delays directly to overseas retail buyers can damage commercial confidence and trigger unwarranted penalties. KnitTrack implements a multi-tier advisory approval workflow:

- **Schema Enforcement:** The ProductionUpdate table includes buyerWarningApproved (Boolean, default false), buyerWarningApprovedBy (String), and buyerWarningApprovedAt (DateTime).
- **Server-Side Visibility Filter:** In backend/src/services/buyer.service.ts, queries serving the Buyer Portal explicitly filter updates with where: { buyerWarningApproved: true }. Unapproved floor warnings (e.g. raw notes with delayRisk: true or critical categories like Machine breakdown or Material shortage) are completely hidden from the buyer payload.
- **Merchandiser Review & Toggle:** Merchandisers and Administrators review floor updates on the Order Detail view. A dedicated action button allows the merchandiser to approve or revoke buyer visibility, toggling buyerWarningApproved via PUT /api/production/updates/:id/approve-warning.
- **Immutable Audit Logging:** Every approval or revocation action writes an entry to the AuditLog table containing the update ID, user ID, action name, and timestamp.
- **Curated Buyer Advisory Presentation:** On the Buyer Portal (frontend/src/pages/BuyerPortalPage.tsx), approved alerts appear under a distinct "Merchandiser-Approved Floor Advisories" card, displaying verified delay reasons, impact explanations, and recovery plans.

17.0 Testing Strategy, Quality Assurance & Verification

The KnitTrack test suite emphasizes automated verification of business logic, mathematical precision, server-side security, offline resilience, and buyer tenant isolation:

Test Suite File	Test Harness	Category / Verified Capabilities	Tests Passed	Pass Rate
delay.test.ts	Vitest 3.2	5-day business week math, actual/required run rate, zero output, past deadline, risk boundary transitions	24 passed	100%
ai.test.ts	Vitest 3.2	Tamil colloquial phrases, English text, regex fallback dictionary, whitespace resilience, audit logging	13 passed	100%
orders.test.ts	Supertest 7.1	Purchase order CRUD, duplicate PO rejection, size ratio JSON validation, supervisor unit scoping	20 passed	100%
auth.test.ts	Supertest 7.1	Credential validation, bcrypt check, HttpOnly cookie issuance, token decoding, logout cookie clearing	12 passed	100%
rbac_isolation.test.ts	Supertest 7.1	Cross-role 403 enforcement, Buyer B vs Buyer C isolation, timeline scoping, supervisor floor limits	13 passed	100%
buyer_approval.test.ts	Supertest 7.1	Merchandiser warning approval, revocation, server-side filtering, buyer modification rejection, idempotency	7 passed	100%
StatusBadge.test.tsx	Vitest 3.2	Frontend semantic badge rendering, risk color classes, accessible screen-reader labels	4 passed	100%
offlineQueue.test.ts	Vitest 3.2	IndexedDB queueing, persistence, auto-sync on online, FIFO order, network retry, clientRequestId idempotency, auth safety	9 passed	100%
GRAND TOTAL	Vitest / Supertest	Complete End-to-End Automated Test Suite Across Full Stack (89 Backend + 13 Frontend)	102 passed	100%

17.1 Production Build & Database Verification

- **Backend TypeScript Compilation:** Executed `npm --prefix backend run build` (tsc). Result: 0 errors, 0 warnings. Clean JavaScript output generated in backend/dist/.
- **Frontend Production Bundle:** Executed `npm --prefix frontend run build` (tsc -b && vite build). Result: 0 errors. Clean client bundle generated in frontend/dist/ in 2.24s.
- **Database Read-Only Verification:** Executed `npx --prefix backend tsx backend/scripts/verify-orders.ts`. Verified 7 seeded orders and 8 user accounts intact in PostgreSQL 17 database.

- **Note on Completed Order C-PO-0512:** While C-PO-0512 carries an ex-factory date of 2026-10-25 (prior to the current reference date), the order status is COMPLETED and all 5,000 planned units across Cutting, Stitching, Finishing, and Packing were 100% completed on or before 2026-10-06. Because remaining quantity is 0 and all stages are marked COMPLETED, the calculation engine (analyseStage) sets remainingQuantity = 0 and projectedDelayDays = 0, correctly classifying the completed order as ON_TRACK.

18.0 Deployment Architecture & Operational Runbook Summary

For step-by-step local setup, environment configurations, and troubleshooting, refer directly to deployment.md. The table below summarizes the environment architectures:

Environment Tier	Frontend Hosting	Backend Service	Database Engine	Object Storage
Local Assessment Dev (Current)	Vite Dev Server (localhost:5173)	Node.js Express (localhost:3001)	PostgreSQL 17 (localhost:5432) via Prisma ORM	Local Disk Storage (backend/uploads/ via Multer)
Cloud Staging (Vercel/Render)	Vercel Edge Network (Global CDN)	Render Web Service (Containerized Node.js)	Neon.tech / Supabase Managed Cloud PostgreSQL	AWS S3 Bucket or Cloudflare R2 with Presigned URLs
High-Volume Enterprise Prod	Vercel / Cloudflare Pages	AWS ECS Fargate (Auto-scaling cluster)	AWS RDS PostgreSQL (Multi-AZ, Read Replicas)	AWS S3 + CloudFront CDN + Redis Session Cache

19.0 Engineering Limitations & Architectural Trade-Offs

To maintain technical credibility, the following constraints reflect deliberate architectural trade-offs made during development:

- **1. Closed-Browser Background Sync:** While offline queueing and automatic sync on reconnection are fully implemented via IndexedDB, syncing while the browser tab is entirely closed requires the ServiceWorker Background Sync API, which remains experimental across iOS WebKit.
- **2. Local Disk vs Cloud Object Storage:** Uploaded stage assembly photos are stored on local filesystem volumes via Multer in development. Production deployment requires migrating to S3-compatible cloud storage with presigned URLs to support stateless container instances.
- **3. In-Memory Rate Limiting:** Express rate limiting currently stores request counts in process memory. In a distributed multi-instance deployment, rate limiting must be backed by a centralized Redis instance.
- **4. Schema Migration Safety Notice:** Prisma db push applies schema changes directly. While non-destructive updates are preserved, destructive changes (e.g. dropping columns

or altering constraints) can result in data loss. Production workflows must use formal migrations and database backups.

20.0 Post-MVP Engineering Roadmap

- **V2 Milestone 1: ServiceWorker Background Sync:** Implement Workbox ServiceWorker background sync for background sync even when the browser tab is dismissed.
- **V2 Milestone 2: IoT Sewing Sensor Integration:** Integrate MQTT industrial telemetry sensors on sewing machine motors to track stitch counts and machine uptime automatically.
- **V2 Milestone 3: Automated PDF Inspection Certificates:** Provide merchandisers with one-click generation of branded PDF Time-and-Action Gantt charts and pre-shipment inspection certificates for international buyers.
- **V2 Milestone 4: WhatsApp Business Shift Reminders:** Automate end-of-shift WhatsApp notifications prompting line supervisors to submit daily outputs directly from messaging channels.

21.0 Architectural Conclusion & Evaluation Baseline

KnitTrack replaces manual, error-prone knitwear tracking practices with an auditable digital operations platform. By combining a mathematically transparent 5-day business week delay engine, an AI-assisted Tamil/English note classifier with robust regex fallback, offline-first IndexedDB queueing with clientRequestId deduplication, server-side RBAC, and curated buyer advisory workflows, the system provides operational visibility across manufacturing stages.

With 102 automated tests passing at a 100% success rate, clean TypeScript compilation across both frontend and backend, and verified data integrity in PostgreSQL 17, KnitTrack represents a stable, thoroughly tested candidate submission for the Isocode Labs software engineering internship assessment.

22.0 References & Verification Artifacts

- **GitHub Project Repository:** https://github.com/ayanali0249/knittrack-production-tracker
- **Deployment Runbook:** deployment.md (included in root repository and submission ZIP)
- **Prisma Database Schema:** backend/prisma/schema.prisma
- **Automated Test Suites:** backend/src/__tests__/ (delay, ai, orders, auth, rbac_isolation, buyer_approval) and frontend/src/__tests__/ (StatusBadge, offlineQueue)
- **Database Verification Script:** backend/scripts/verify-orders.ts (safe read-only query harness)